Lifting and Shoring – General Data

1. Information and Warnings

A. Refer to PSM 1-6-2T for storage, maintenance and inspection requirements.

B. The slings and lifting devices are designed in accordance with ASME B30.20 “Safety Standard for Below-the Hook Lifting Devices”. It is the responsibility of the owner/user to inspect, maintain and use the lifting device in accordance with ASME B30.20 “Safety Standard for Below-the-Hook Lifting Devices”, PSM 1-6-2T and PSM 1-64-2. It is the responsibility of the owner/user to ensure compliance with all standards if the sling or lifting device is used in conjunction with other lifting systems, such as, but not limited to ASME B30.16 “Safety Standards for Overhead Hoists (Underhung)”.

C. Modifications to slings and lifting devices are not permitted unless approved by Viking.

D. Viking slings and lifting devices are recommended for indoor use, or outdoor use when the following conditions are met:

- Adequate sunlight or supplementary light is present
- Winds do not exceed 3 kts (5.6 km/h)
- Precipitation is not present

E. Viking slings and lifting devices must be used to lift loads vertically only.

F. Slings and lifting devices are not recommended for transporting a suspended load.

2. Approximate Weight of Components

Table 1 Approximate Weight of Components

ITEM	APPROXIMATE WEIGHT (POUNDS)
Engine	324
Propeller	134
Wing	910
Wing Strut	90
Aileron	79
Flaps	133
Vertical Stabilizer	53
Rudder	67
Horizontal Stabilizer	94
Elevator	70

3. Hoisting the Aircraft

CAUTION SLING AND LIFTING DEVICE OPERATORS ARE RESPONSIBLE FOR COMPLIANCE WITH ALL APPLICABLE INDUSTRY STANDARDS, AND WITH ALL APPLICABLE LOCAL REGULATIONS.

A. Prior to carrying out a lift, the sling or lifting device must be inspected in accordance with inspection requirements given in PSM 1-6-2T, Chapter 7.

CAUTION **A REGULAR SCHEDULE OF INSPECTION OF SLINGS AND LIFTING DEVICES IN ACCORDANCE WITH THE REQUIREMENTS OF ASME B30.20 SHALL BE ESTABLISHED AND RECORDS MAINTAINED.**

B. The entire aircraft may be hoisted by using an approved aircraft sling or lifting device (refer to PSM 1-6-2T).

C. Read and follow all information in Para 1.

WARNING **THE SLINGS AND LIFTING DEVICES ARE NOT DESIGNED FOR, AND MUST NOT BE USED FOR LIFTING, SUPPORTING, OR TRANSPORTATION OF PERSONNEL.**

D. Remove any cargo, loose items, passengers and personnel from aircraft prior to hoisting. The aircraft may be lifted with interior fixtures.

E. It is recommended that the aircraft be defueled.

F. Confirm aircraft weight is within limitation of the sling or lifting device.

WARNING **THE SLINGS AND LIFTING DEVICES MUST ONLY BE USED TO LIFT THE DHC-6 COMPONENT THAT THEY ARE INTENDED FOR AND MUST NOT BE USED TO LIFT ANY PARTS, ASSEMBLIES OR AIRCRAFT OTHER THAN THOSE DESCRIBED WITHIN THIS MANUAL. IF AN INCORRECT SLING OR LIFTING DEVICE IS USED, FAILURE OF THE SLING OR LIFTING DEVICE MAY OCCUR AND INJURE PERSONNEL OR CAUSE DAMAGE TO AIRCRAFT OR EQUIPMENT.**

G. Attach front sling attachment brackets in accordance with Figure 1, Detail B. Insert the pin through the bracket and the forward wing attachment bolt. Confirm pin is locked in place.

Note The bushings are not required when the aircraft is lifted with wings attached.

H. Attach rear sling attachment brackets in accordance with Figure 1, Detail A. Remove the shackle, to allow the bracket to open in a scissor fashion to fit over the aft wing attachment bolt. Close the bracket and re-insert the shackle to lock the bracket from opening.

Note The bushings are not required when the aircraft is lifted with wings attached.

I. Attach master link to a lifting crane with the applicable lifting capacity.

J. Connect front and rear sling attachment brackets, installed in Para 3.G and Para 3.H, to lifting device lower chains.

K. Confirm attachments are secure and personnel are clear.

CAUTION **SLING AND LIFTING DEVICE OPERATORS MUST BE TRAINED IN OPERATION OF ANY EQUIPMENT HE OR SHE IS OPERATING.**

L. Carry out lift in accordance with all applicable local regulations.

4. Hoisting the Fuselage

A. Read and follow all information in Para 1.

B. The fuselage cannot be lifted without the wings if the empennage is installed (refer to Figure 1).

C. Remove any cargo, loose items, passengers and personnel from aircraft prior to hoisting. The fuselage may be lifted with interior fixtures.

D. Carry out lift as described in Para 3.F through Para 3.L.

5. Hoisting a Wing

Note Sling and lifting device operators are responsible for compliance with all applicable industry standards, and with all applicable local regulations.

Note Sling and lifting device operators must be trained in operation of any equipment they are operating.

A. Prior to carrying out a lift, the sling or lifting device must be inspected in accordance with inspection requirements given in PSM 1-6-2T, Chapter 7.

CAUTION **A REGULAR SCHEDULE OF INSPECTION OF SLINGS AND LIFTING DEVICES IN ACCORDANCE WITH THE REQUIREMENTS OF ASME B30.20 SHALL BE ESTABLISHED AND RECORDS MAINTAINED.**

B. The wing may be hoisted by using an approved wing sling or lifting device (refer to PSM 1-6-2T).

C. Read and follow all information in Para 1. of this section.

WARNING **THE SLINGS AND LIFTING DEVICES ARE NOT DESIGNED FOR LIFTING, SUPPORTING OR TRANSPORTATION OF PERSONNEL, AND MUST NOT BE USED FOR THOSE PURPOSES.**

D. Confirm that the total weight of the wing (refer to Table 1 — Approximate Weight of Components), the lifting device, any rigging hardware, and any other additional weight being lifted is within limitations of the sling or lifting device, and the crane being used to carry out the lift.

CAUTION **THE LONG-RANGE FUEL SYSTEM WING TANKS (IF EQUIPPED) MUST BE DEFUELED, AND THE ENGINE MUST BE REMOVED.**

WARNING **THE SLINGS AND LIFTING DEVICES MUST ONLY BE USED TO LIFT THE DHC-6 COMPONENT THAT THEY ARE INTENDED FOR. THEY MUST NOT BE USED TO LIFT ANY PARTS, ASSEMBLIES, OR AIRCRAFT OTHER THAN THOSE DESCRIBED WITHIN THIS MANUAL.**

E. Remove Wing

Refer to Figure 1 and 57-00-00, Wings – General Data for details.

1) Disconnect the wing sling cradles from the spreader.

2) Attach the spreader lifting eye to the crane and suspend over the wing.

3) Place one wing cradle below the wing at approximately wing station 137.

4) Place the remaining wing cradle below the wing at approximately wing station 209.

5) Attach the cables to the cradles using the quick-release ball-lock pins. Select the appropriate attachment, depending on whether the wing is being lifted with or without flaps and ailerons.

6) Lift crane until slight tension is applied to all cables.

7) Confirm all attachments are secure.

8) Lift crane until full weight of wing is supported by the lifting sling.

9) Remove wing. Refer to 57-00-00, Wings – General Data.

10) Check the balance of the wing. If the wing is not balanced, a 30-40 lb ballast weight may be placed on top of the wing near the wing tip. Balance can be adjusted by moving the ballast weight inboard or outboard. Ensure the weight does not damage the wing skin.

11) Confirm personnel are clear and carry out lift in accordance with all applicable local regulations.

12) Lower wing on to a suitable support surface.

13) When lift is complete and wing is securely supported, remove the sling from the wing.

F. Install Wing

Refer to Figure 1 and 57-00-00, Wings – General Data for details.

1) Disconnect the wing sling cradles from the spreader.

2) Attach the spreader lifting eye to the crane and suspend over the wing.

3) Place one wing cradle below the wing at approximately wing station 137.

4) Place the remaining wing cradle below the wing at approximately wing station 209.

5) Attach the cables to the cradles using the quick-release ball-lock pins. Select the appropriate attachment, depending on whether the wing is being lifted with or without flaps and ailerons.

6) Lift crane until slight tension is applied to all cables.

7) Confirm all attachments are secure.

8) Lift crane until full weight of wing is supported by the lifting sling.

9) Check the balance of the wing. If the wing is not balanced, a 30-40 lb ballast weight may be placed on top of the wing near the wing tip. Balance can be adjusted by moving the ballast weight inboard or outboard. Ensure the weight does not damage the wing skin.

10) Confirm personnel are clear and carry out lift in accordance with all applicable local regulations.

11) Move wing to the aircraft and position in place.

12) Install wing. Refer to 57-00-00, Wings – General Data.

13) When lift is complete and wing is secured to the aircraft, remove the sling from the wing.

6. Hoisting the Vertical Stabilizer

Note Sling and lifting device operators are responsible for compliance with all applicable industry standards, and with all applicable local regulations.

Note Sling and lifting device operators must be trained in operation of any equipment they are operating.

A. Prior to carrying out a lift, the sling or lifting device must be inspected in accordance with inspection requirements given in PSM 1-6-2T, Chapter 7.

CAUTION A REGULAR SCHEDULE OF INSPECTION OF SLINGS AND LIFTING DEVICES IN ACCORDANCE WITH THE REQUIREMENTS OF ASME B30.20 SHALL BE ESTABLISHED AND RECORDS MAINTAINED.

B. The vertical stabilizer may be hoisted by using an approved sling or lifting device (refer to PSM 1-6-2T).

C. Read and follow all information in Para 1. of this section.

WARNING **THE SLINGS AND LIFTING DEVICES ARE NOT DESIGNED FOR LIFTING, SUPPORTING OR TRANSPORTATION OF PERSONNEL, AND MUST NOT BE USED FOR THOSE PURPOSES.**

D. Confirm that the total weight of the vertical stabilizer (refer to Table 1 — Approximate Weight of Components), the lifting device, any rigging hardware, and any other additional weight being lifted is within limitations of the sling or lifting device, and the crane being used to carry out the lift.

WARNING **THE SLINGS AND LIFTING DEVICES MUST ONLY BE USED TO LIFT THE DHC-6 COMPONENT THAT THEY ARE INTENDED FOR. THEY MUST NOT BE USED TO LIFT ANY PARTS, ASSEMBLIES, OR AIRCRAFT OTHER THAN THOSE DESCRIBED WITHIN THIS MANUAL.**

E. Remove Vertical Stabilizer

Refer to Figure 1 and 55-30-00, Vertical Stabilizer – General Data for details.

1) Remove and retain AN525-10R10 screws and AN315-3 nuts supplied in thimble of the sling.

2) Attach the cables to the vertical stabilizer using the supplied AN525-10R10 screws. Do not use the AN315-3 nuts while lifting.

3) Confirm the sling can rotate freely about the attachments.

4) Suspend crane above the vertical stabilizer and attach the sling to the crane.

5) Lift crane until slight tension is applied to all cables.

6) Confirm all shackles, ball-lock pins, and other attachments are secure.

7) Remove vertical stabilizer. Refer to 55-30-00, Vertical Stabilizer – General Data.

8) Confirm personnel are clear and carry out lift in accordance with all applicable local regulations.

9) Lower vertical stabilizer on to a suitable surface.

10) When lift is complete and vertical stabilizer is securely supported, remove the sling from the vertical stabilizer.

11) Re-install the AN525-10R10 screws into the thimbles of the sling and retain screws with AN315-3 nuts.

F. Install Vertical Stabilizer

Refer to Figure 1 and 55-30-00, Vertical Stabilizer – General Data for details.

1) Remove and retain AN525-10R10 screws and AN315-3 nuts supplied in thimble of the sling.

2) Attach the cables to the vertical stabilizer using the supplied AN525-10R10 screws. Do not use the AN315-3 nuts while lifting.

3) Confirm the sling can rotate freely about the attachments. (Refer to Figure 1).

4) Suspend crane above the vertical stabilizer and attach the sling to the crane.

5) Lift crane until slight tension is applied to all cables.

6) Confirm all shackles, ball-lock pins, and other attachments are secure.

7) Confirm personnel are clear and carry out lift in accordance with all applicable local regulations, allowing vertical stabilizer to rotate on sling attachments.

8) Move vertical stabilizer to the aircraft and position in place.

9) Install vertical stabilizer. Refer to 55-30-00, Vertical Stabilizer – General Data.

10) When lift is complete and vertical stabilizer is secured to the aircraft, remove the sling from the vertical stabilizer.

11) Re-install the AN525-10R10 screws into the thimbles of the sling and retain screws with AN315-3 nuts.

7. Hoisting the Horizontal Stabilizer

Note Sling and lifting device operators are responsible for compliance with all applicable industry standards, and with all applicable local regulations.

Note Sling and lifting device operators must be trained in operation of any equipment they are operating.

A. Prior to carrying out a lift, the sling or lifting device must be inspected in accordance with inspection requirements given in PSM 1-6-2T, Chapter 7.

CAUTION A REGULAR SCHEDULE OF INSPECTION OF SLINGS AND LIFTING DEVICES IN ACCORDANCE WITH THE REQUIREMENTS OF ASME B30.20 SHALL BE ESTABLISHED AND RECORDS MAINTAINED.

B. The horizontal stabilizer may be hoisted by using an approved sling or lifting device (refer to PSM 1-6-2T).

C. Read and follow all information in Para 1. of this section.

WARNING THE SLINGS AND LIFTING DEVICES ARE NOT DESIGNED FOR LIFTING, SUPPORTING OR TRANSPORTATION OF PERSONNEL, AND MUST NOT BE USED FOR THOSE PURPOSES.

D. Confirm that the total weight of the horizontal stabilizer (refer to Table 1 — Approximate Weight of Components), the lifting device, any rigging hardware, and any other additional weight being lifted is within limitations of the sling or lifting device, and the crane being used to carry out the lift.

WARNING THE SLINGS AND LIFTING DEVICES MUST ONLY BE USED TO LIFT THE DHC-6 COMPONENT THAT THEY ARE INTENDED FOR. THEY MUST NOT BE USED TO LIFT ANY PARTS, ASSEMBLIES, OR AIRCRAFT OTHER THAN THOSE DESCRIBED WITHIN THIS MANUAL.

E. Remove Horizontal Stabilizer

Refer to Figure 1 and 55-10-00, Horizontal Stabilizer – General Data for details.

1) Locate each of the forward lugs of the sling on the forward upper surface of the horizontal stabilizer and install the supplied bolts as illustrated in Figure 1.

2) Select the appropriate cable for attachment to the aft lifting point of the horizontal stabilizer. Use the short cable to lift the horizontal stabilizer with the elevator attached, or use the long cable to lift the horizontal stabilizer without the elevator attached.

3) Place the shackle of the aft cable over the lug located above the aft spar of the horizontal stabilizer and install the quick-release ball-lock pin attached to the sling.

4) Suspend crane above the horizontal stabilizer and attach the lifting ring to the crane.

5) Lift crane until slight tension is applied to all cables.

6) Confirm all attachments are secure.

7) Remove horizontal stabilizer. Refer to 55-10-00, Horizontal Stabilizer – General Data.

8) Confirm personnel are clear and carry out lift in accordance with all applicable local regulations.

9) Lower horizontal stabilizer on to a suitable surface.

10) When lift is complete and horizontal stabilizer is securely supported, remove the sling from the horizontal stabilizer.

F. Install Horizontal Stabilizer

Refer to Figure 1 and 55-10-00, Horizontal Stabilizer – General Data for details.

1) Locate each of the forward lugs of the sling on the forward upper surface of the horizontal stabilizer and install the supplied bolts as illustrated in Figure 1.

2) Select the appropriate cable for attachment to the aft lifting point of the horizontal stabilizer. Use the short cable to lift the horizontal stabilizer with the elevator attached, or use the long cable to lift the horizontal stabilizer without the elevator attached.

3) Place the shackle of the aft cable over the lug located above the aft spar of the horizontal stabilizer and install the quick-release ball-lock pin attached to the sling.

4) Suspend crane above the horizontal stabilizer and attach the lifting ring to the crane.

5) Lift crane until slight tension is applied to all cables.

6) Confirm all attachments are secure.

7) Confirm personnel are clear and carry out lift in accordance with all applicable local regulations.

8) Move horizontal stabilizer to the aircraft and position in place.

9) Install horizontal stabilizer. Refer to 55-10-00, Horizontal Stabilizer – General Data.

10) When lift is complete and horizontal stabilizer is secured to the aircraft, remove the sling from the horizontal stabilizer.

8. Hoisting the Rudder

Note Sling and lifting device operators are responsible for compliance with all applicable industry standards, and with all applicable local regulations.

Note Sling and lifting device operators must be trained in operation of any equipment they are operating.

A. Prior to carrying out a lift, the sling or lifting device must be inspected in accordance with inspection requirements given in PSM 1-6-2T, Chapter 7.

CAUTION **A REGULAR SCHEDULE OF INSPECTION OF SLINGS AND LIFTING DEVICES IN ACCORDANCE WITH THE REQUIREMENTS OF ASME B30.20 SHALL BE ESTABLISHED AND RECORDS MAINTAINED.**

B. The rudder may be hoisted by using an approved sling or lifting device (refer to PSM 1-6-2T).

C. Read and follow all information in Para 1. of this section.

WARNING **THE SLINGS AND LIFTING DEVICES ARE NOT DESIGNED FOR LIFTING, SUPPORTING OR TRANSPORTATION OF PERSONNEL, AND MUST NOT BE USED FOR THOSE PURPOSES.**

D. Confirm that the total weight of the rudder (refer to Table 1 — Approximate Weight of Components), the lifting device, any rigging hardware, and any other additional weight being lifted is within limitations of the sling or lifting device, and the crane being used to carry out the lift.

WARNING **THE SLINGS AND LIFTING DEVICES MUST ONLY BE USED TO LIFT THE DHC-6 COMPONENT THAT THEY ARE INTENDED FOR. THEY MUST NOT BE USED TO LIFT ANY PARTS, ASSEMBLIES, OR AIRCRAFT OTHER THAN THOSE DESCRIBED WITHIN THIS MANUAL.**

E. Remove Rudder

Refer to Figure 1 and 27-20-11, Rudder – Maintenance Practices for details.

1) Attach the links of sling to the top of the rudder using the supplied hardware. Ensure that the Teflon rubbing plate faces the rudder skin.

2) Confirm the sling can rotate freely about the attachments.

3) Suspend crane above the rudder and attach the sling to the crane.

4) Lift crane until slight tension is applied to all cables.

5) Confirm all attachments are secure.

6) Remove rudder. (Refer to 27-20-11, Rudder – Maintenance Practices).

7) Confirm personnel are clear and carry out lift in accordance with all applicable local regulations.

8) Lower rudder on to a suitable surface and rotate manually, as required, to prevent damage, ensuring personnel do not position themselves under the suspended load.

9) When lift is complete and rudder is securely supported, remove the sling from the rudder.

F. Install Rudder

Refer to Figure 1 and 27-20-11, Rudder – Maintenance Practices for details.

1) Attach the links of sling to the top of the rudder using the supplied hardware. Ensure that the Teflon rubbing plate faces the rudder skin.

2) Confirm the sling can rotate freely about the attachments.

3) Suspend crane above the rudder and attach the sling to the crane.

4) Lift crane until slight tension is applied to all cables.

5) Confirm all attachments are secure.

6) Confirm personnel are clear and carry out lift in accordance with all applicable local regulations, allowing rudder to rotate on sling attachments.

7) Move rudder to the aircraft and position in place.

8) Install rudder. (Refer to 27-20-11, Rudder – Maintenance Practices).

9) When lift is complete and rudder is secured to the aircraft, remove the sling from the rudder.

9. Hoisting an Engine

The engine may be hoisted using the engine sling (SD12541) attached to the lifting points of the engine. If desired, the engine may be removed or installed with propeller attached (refer to Figure 2).

10. Hoisting a Propeller

A. From a vertical position:

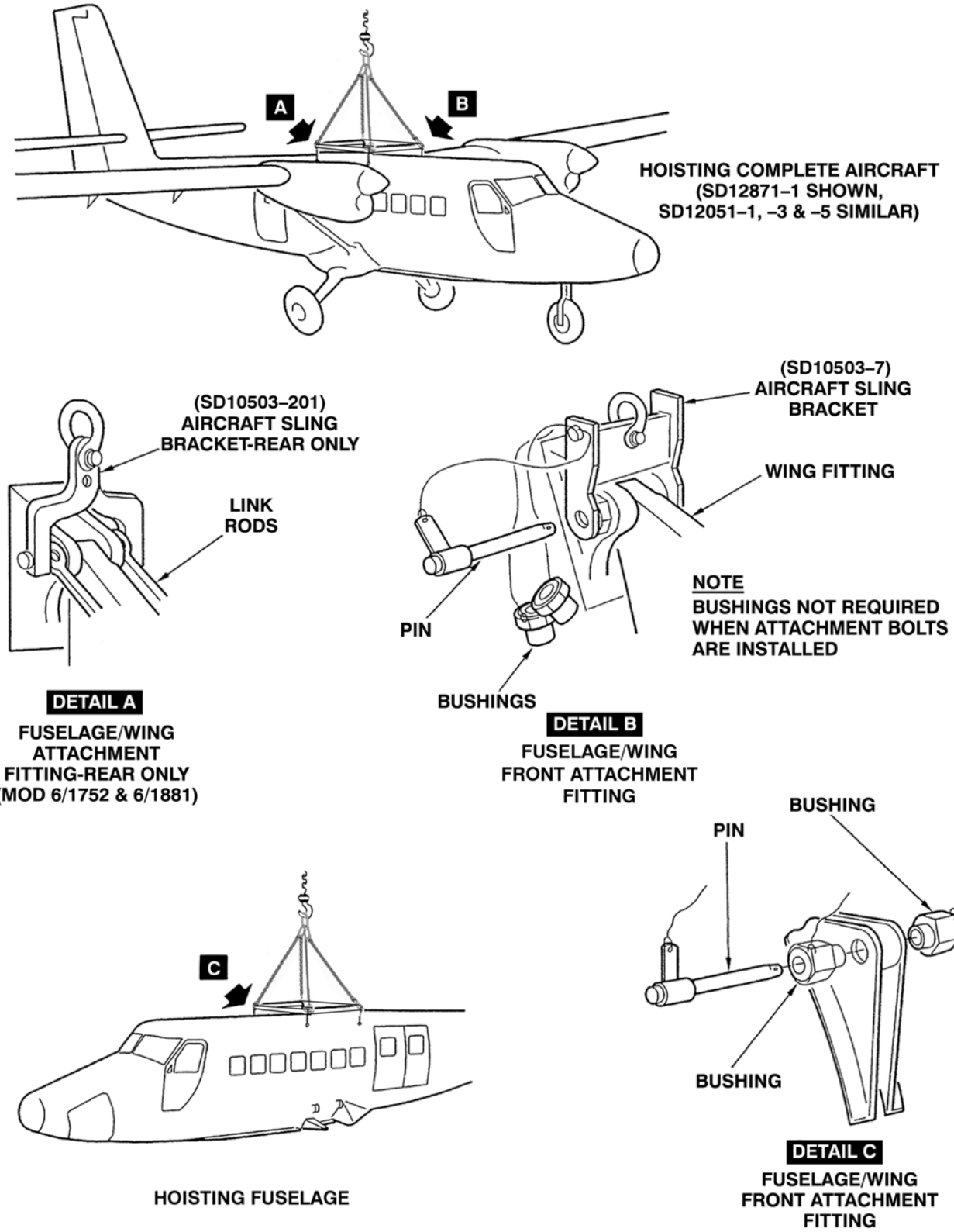
Position the propeller with one blade pointing vertically down, loop sling (SD5534) around roots of upper blades and hoist using a crane of sufficient capacity (refer to Figure 2).

B. From a horizontal position:

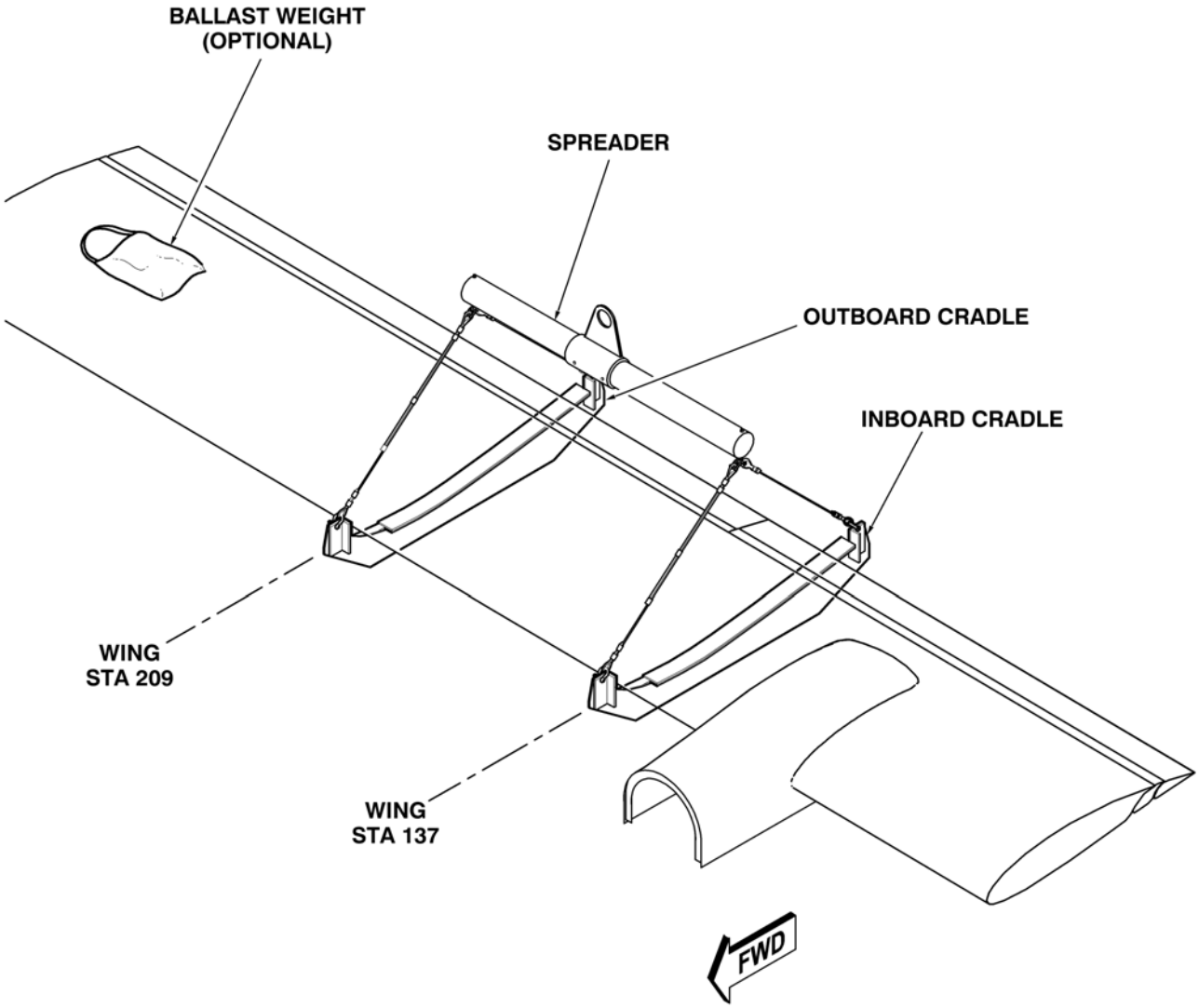
Loop sling (SD5534) around roots of two blades, support unslung blade to prevent damage to tip, and hoist using a crane of sufficient capacity.

11. Jacking the Aircraft

Five jacking points are provided on the aircraft and are located on the underside of the fuselage and main gear. Each jacking pad is coded with a letter, the related insert shows the jacking application. Tail lifting or slinging is accomplished by a bar inserted in the rear fuselage lifting tube and supported by two jacks, one at each end (refer to Figure 3).

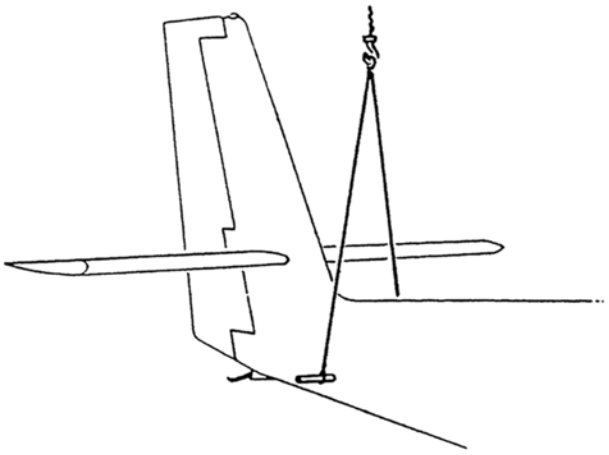


Aircraft Hoisting
Figure 1 (Sheet 1 of 4)

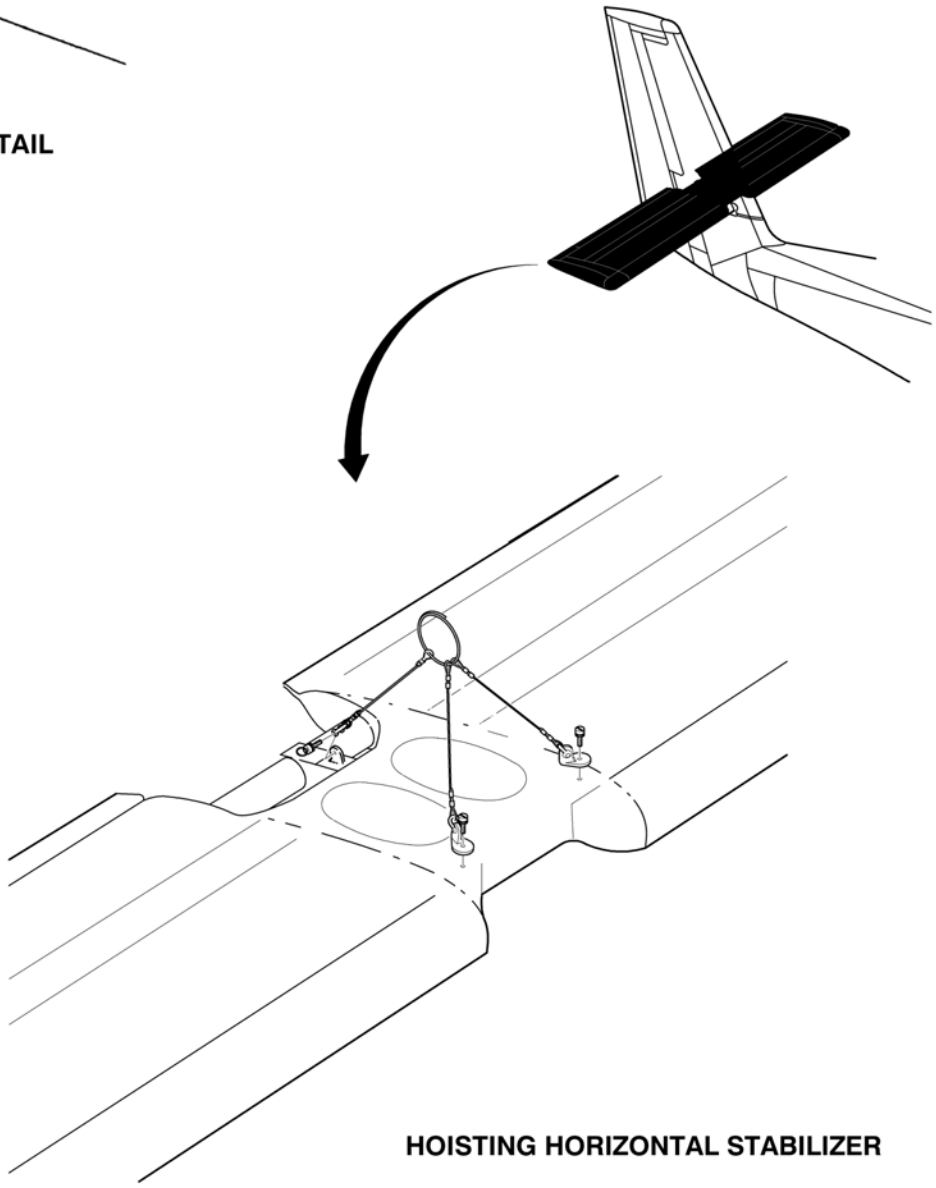


HOISTING WING

Aircraft Hoisting
Figure 1 (Sheet 2 of 4)

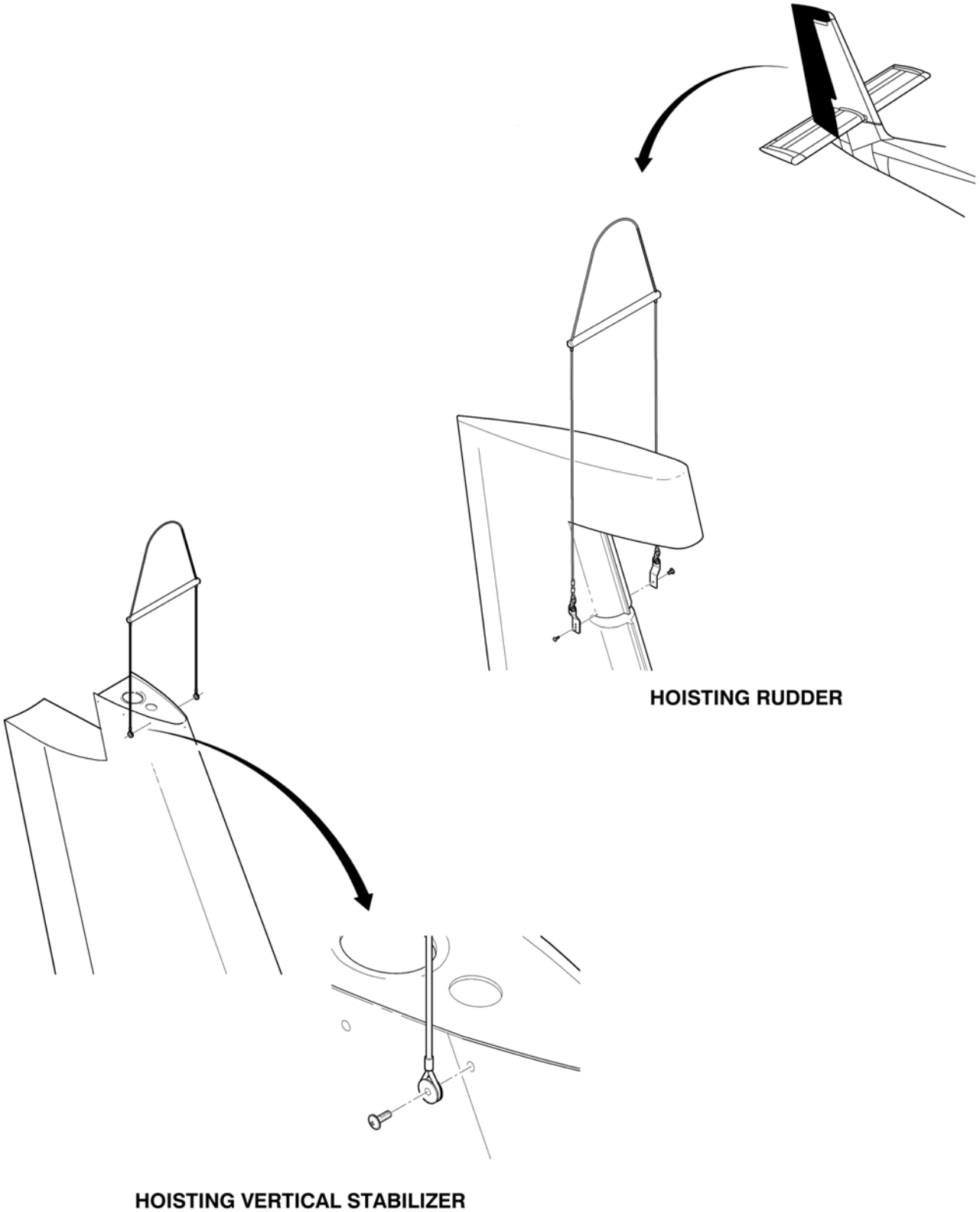


HOISTING TAIL

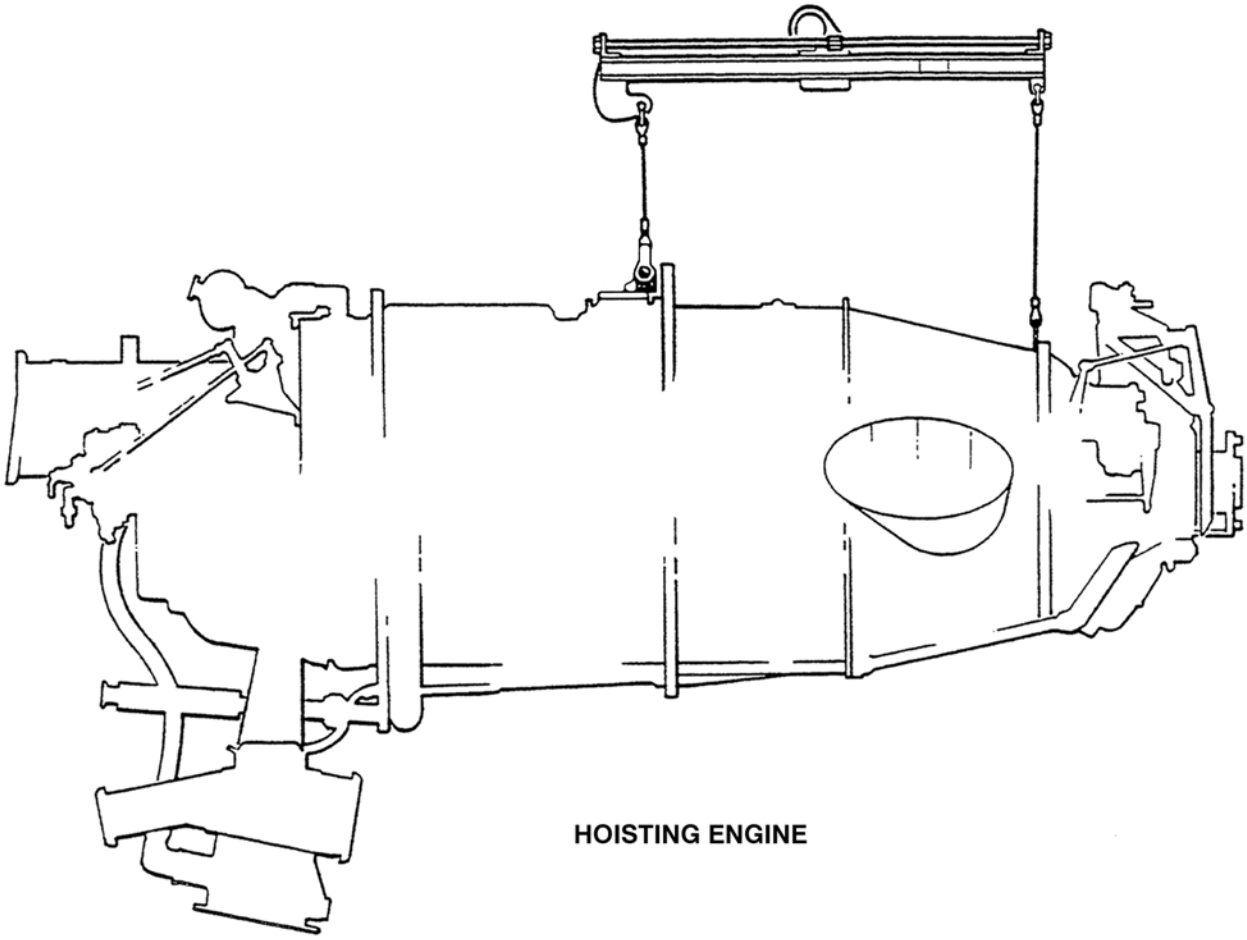


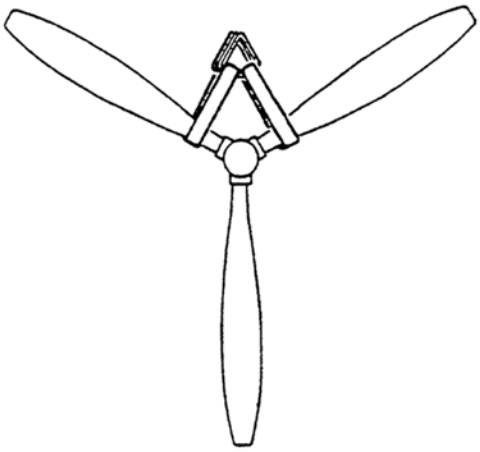
HOISTING HORIZONTAL STABILIZER

Aircraft Hoisting
Figure 1 (Sheet 3 of 4)

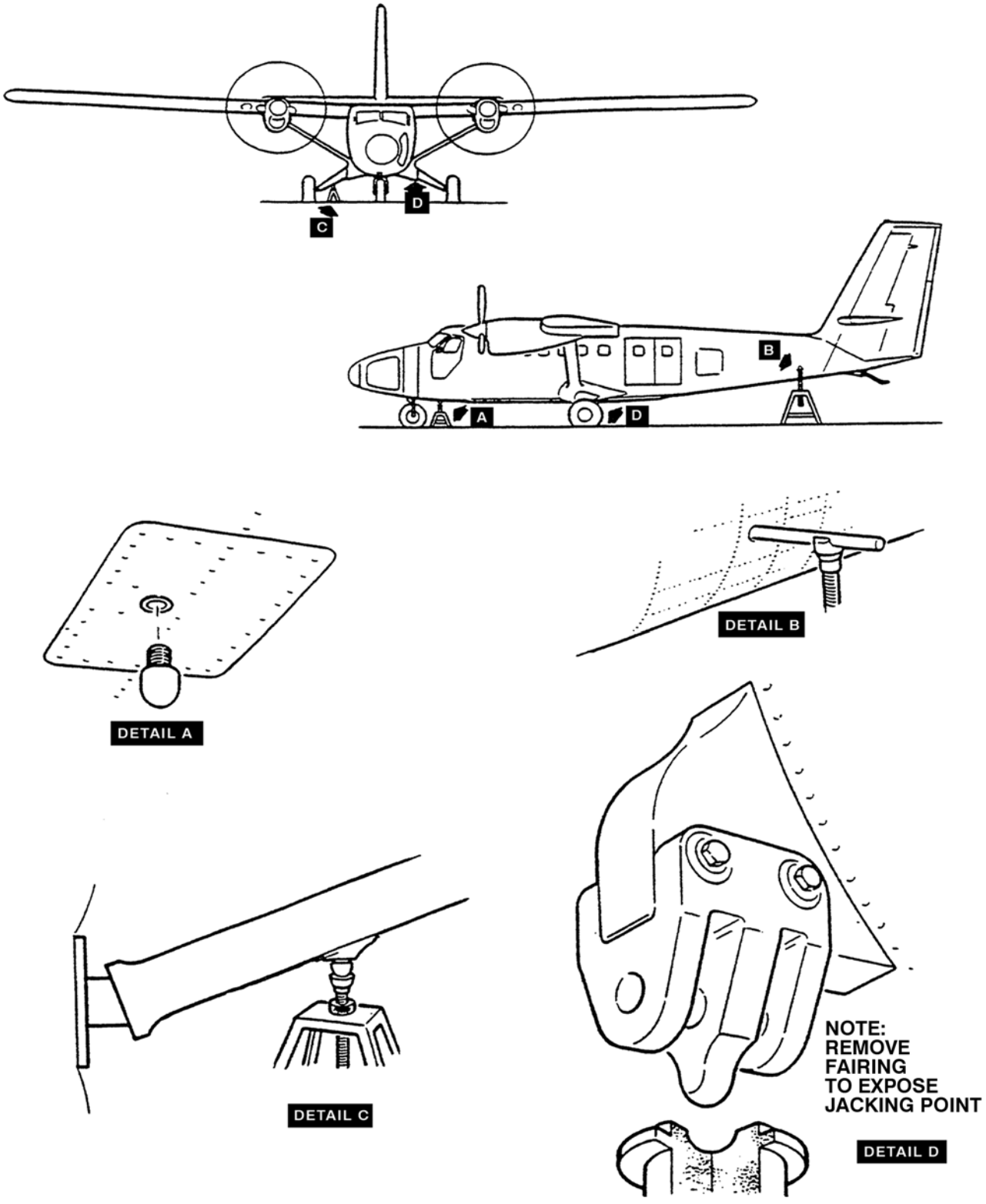


Aircraft Hoisting
Figure 1 (Sheet 4 of 4)





Engine and Propeller Hoisting
Figure 2



Aircraft Jacking
Figure 3

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